(Please write your Exam Roll No.) *Exam Roll No.....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JUNE 2024

Paper Code:CIE-332T/IOT-330T ***Subject- Programming in Python***

Time: 3hours ***Max. Marks: 75***

Note : Attempt five questions in all including Q. No. 1 which is compulsory. Select one Question from each Unit. Assume suitable missing data if any.

Q1 Answer following in brief. **[5x5 = 25]**

(a) What is the role of data frame? How can data frames be merged?
(b) What are the features of Python Programming language?
(c) Differentiate between list, tuple and dictionary with example.
(d) Write a short note on different methods to read data from a file.
(e) How is exception handling implemented in python?

UNIT-I

Q2 (a) What are loop interruption statements? What is the difference between break, continue and pass? 17. Write a program that prints all integers lying between 1 and 50 that aren't divisible by either 2 or 3. **[6]**

(b) Explain the different numeric, assignment, augmented operators used in python. **[3]**

(c) Explain how operator precedence is used in evaluating expressions. **[3.5]**

Q3 (a) What are the commonly used operators in python? Explain operator overloading with the help of an example. **[6]**

(b) What is type conversion? Write a python program to demonstrate number type conversions. **[6.5]**

UNIT-II

Q4 (a) What are the different ways of string manipulation in python? WAP in python check whether the string is Symmetrical or Palindrome **[6]**

(b) What are functions in python? WAP in python to use Pythagoras theorem to calculate the length of the hypotenuse of a right-angled triangle where the other sides have lengths 3 and 4. **[6.5]**

Q5 (a) Define the following: find(), isalnum(), zfill(), strip(), split() and rindex(). **[6]**

(b) Write the definition of a function Reverse(X). WAP to display the elements in reverse order such that each displayed element is twice of the original element (element *2) of the List X. **[3]**

(c) Write a function mul ( ) which accepts list L, odd_L, even_L as its parameters. Transfer all even values of list L to even_L and all odd values of L to odd_L. **[3.5]**

UNIT-III

Q6 (a) Write a program that accepts filename as an input from the user. Open the file and count the number of times a character appears in the file. **[6]**

(b) What are files? Why do we need them? What are different access modes in which you can open file? **[6.5]**

Q7 (a) What is directory access? How do you walk through a directory tree? **[6]**

(b) Write a python program to merge two files and then change the content of this file to upper case letters. **[6.5]**

P.T.O.